

FinanceGuard

Bank Churn Prediction — Model Comparison Report

Integrative Project M4 · Soy Henry Data Science · Daniel Palmera

1. EDA + Logistic Regression (Baseline)

FinanceGuard has a **20.4% churn rate** — 2,037 out of 10,000 clients. Estimated annual loss: **\$4,074,000**. The class imbalance ratio is 1:3.9. Logistic Regression was trained as baseline with `class_weight='balanced'`, `StandardScaler`, and `StratifiedKFold` (5 folds).

Metric	Value	Interpretation
Accuracy	0.7135	Not primary metric — imbalanced data
Precision (Churn)	0.3872	Only 4 of 10 alerts are correct
Recall (Churn)	0.7002	Detects 7 of 10 real churners
F1-Score (Churn)	0.4987	Precision-recall balance
ROC-AUC	0.7771	Baseline discriminative capacity
PR-AUC	~0.52	More informative for minority class

Confusion Matrix: TN=1,142 | FP=451 | FN=122 | TP=285

Key Odds Ratios: Age=2.23 · Geo_Germany=2.04 · IsActiveMember=0.41 (protective factor)

2. Gradient Boosting — Model Comparison

8 models trained and compared. All advanced models outperform the baseline. Grid Search (270 fits) and Optuna (50 trials) used for hyperparameter optimization.

Model	ROC-AUC	PR-AUC	F1	Recall
Stacking (XGB+LGB+CAT)	0.8709 ★	0.7209	0.6139	0.7813
XGBoost (Optuna)	0.8700	0.7182	0.6062	0.7641
XGBoost (Grid Search)	0.8694	0.7191	0.6077	0.7592
CatBoost	0.8689	0.7213	0.6195	0.7420
Random Forest	0.8633	0.6981	0.6155	0.7101
XGBoost (early stop)	0.8624	0.7084	0.6079	0.7543
LightGBM	0.8569	0.6938	0.6150	0.6929
Logistic Reg. (Baseline)	0.7771	0.4678	0.4987	0.7002

Best Grid Search config: lr=0.05, max_depth=4, n_estimators=200, CV-AUC=0.8658

Best Optuna trial: n_estimators=257, max_depth=3, lr=0.0386, AUC=0.8700

3. Customer Segmentation — K-Means + DBSCAN

PCA (10 components, 90% variance) applied before K-Means. K=3 selected by Elbow Method (WCSS) and Silhouette Score. DBSCAN (eps=1.5, min_samples=20) for outlier detection.

Segment	Clients	Avg Balance	Churn Rate	Est. Annual Loss	Priority
0	2,477	\$61,818	16.7%	\$826,000	Medium
1	5,014	\$62,093	16.2%	\$1,620,000	Medium
2	2,509	\$119,730	32.4%	\$1,628,000	HIGH
DBSCAN Outliers	4	Variable	100%	Variable	CRITICAL

K-Means Silhouette: 0.2022 | PCA+K-Means Silhouette: 0.2055 (+1.6%)

Estimated savings with personalized strategy vs. mass campaign: \$796,300/year

4. Threshold Optimization (Extra Credit)

The default threshold (0.50) generates net losses because the asymmetric cost of errors is not considered: FN=\$2,000 loss vs FP=\$100 cost.

Threshold	Recall	TP	FP	Net Benefit
0.50 (default)	0.700	285	451	-\$132,100
0.11 (optimal)	0.936	381	923	+\$46,200
0.627 (best F1)	lower	fewer	fewer	-\$202,400

Improvement vs. default: +\$178,300 per campaign cycle | +\$713,200/year (4 cycles)

5. Final Conclusions

Finding	Value	Business Impact
Best model	Stacking AUC=0.8709	+12.1% vs baseline
Optimal threshold	0.11 (Recall=93.6%)	+\$178,300/campaign cycle
Critical segment	Seg. 2: \$119,730 bal., 32.4% churn	Immediate priority intervention
DBSCAN outliers	4 clients, 100% churn	Immediate manual attention
Derived features	cluster_label, churn_risk_cluster	Can further improve AUC

Recommendation: Deploy the Stacking model with threshold=0.11 for churn retention campaigns, combined with the 3-segment strategy for personalized interventions.